

Lightweight Deep Learning Framework for Fluorescent Neuronal Cell Counting from Microscopy Images

Impact Statement

Automated analysis of microscopy images is becoming increasingly important in biomedical and neuroscience research. Many existing cell counting methods rely on complex segmentation pipelines that require detailed pixel-level annotations and computationally expensive models. Such approaches can be difficult to deploy in environments with limited computational resources.

This work presents a lightweight deep learning framework for fluorescent neuronal cell counting that directly predicts cell counts from microscopy images without requiring segmentation during inference. The proposed approach simplifies the overall analysis pipeline while still producing reliable counting estimates.

The study demonstrates that compact convolutional models can achieve practical counting performance with relatively low computational cost. This makes the proposed framework suitable for rapid analysis workflows and resource-constrained environments where large segmentation networks may not be practical.

The findings also support the broader idea that direct feature-based counting can serve as an effective alternative to segmentation-heavy methods for certain biomedical imaging tasks.